



LNG Trading Spreads

Is the US-to-Europe and Asia-vs-Europe arbitrage open?

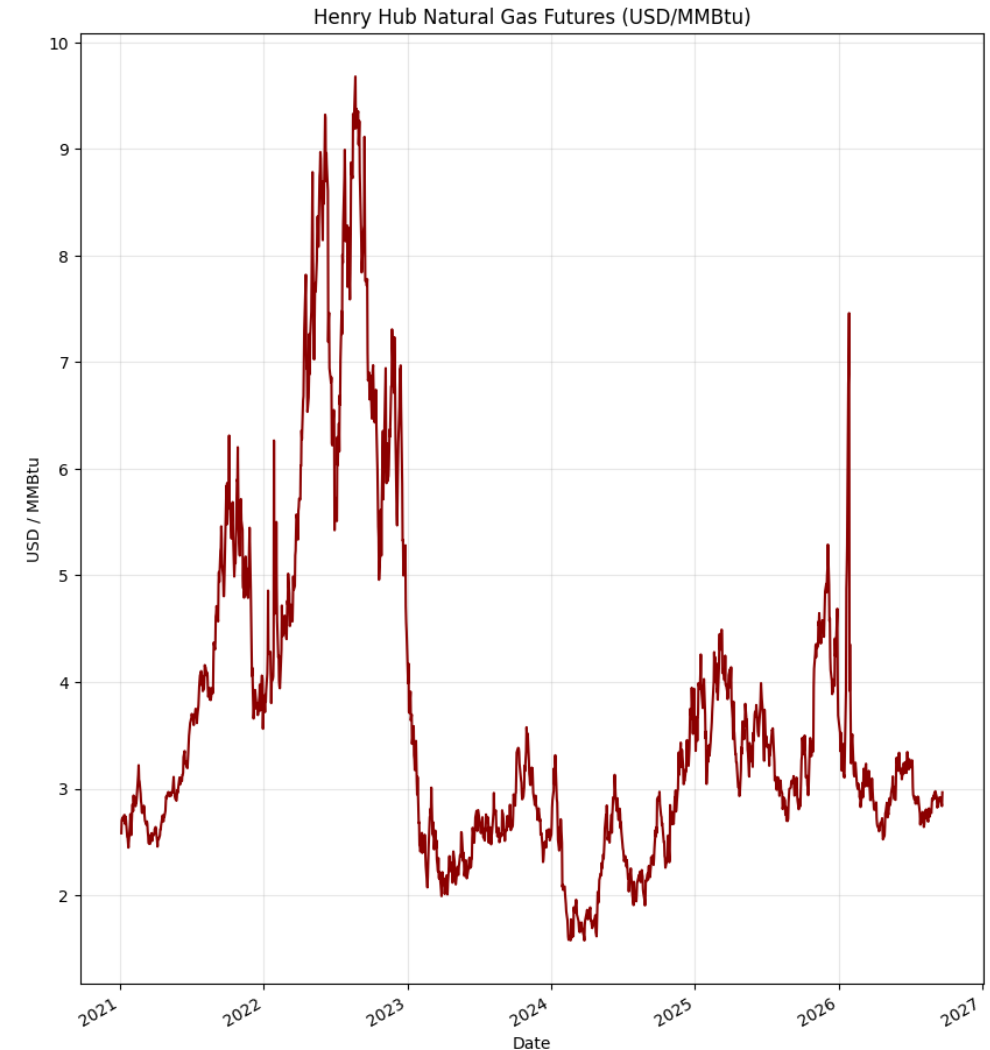
Henry Hub vs TTF vs JKM | Daily data, Jan 2021 - Sep 2026

Approach

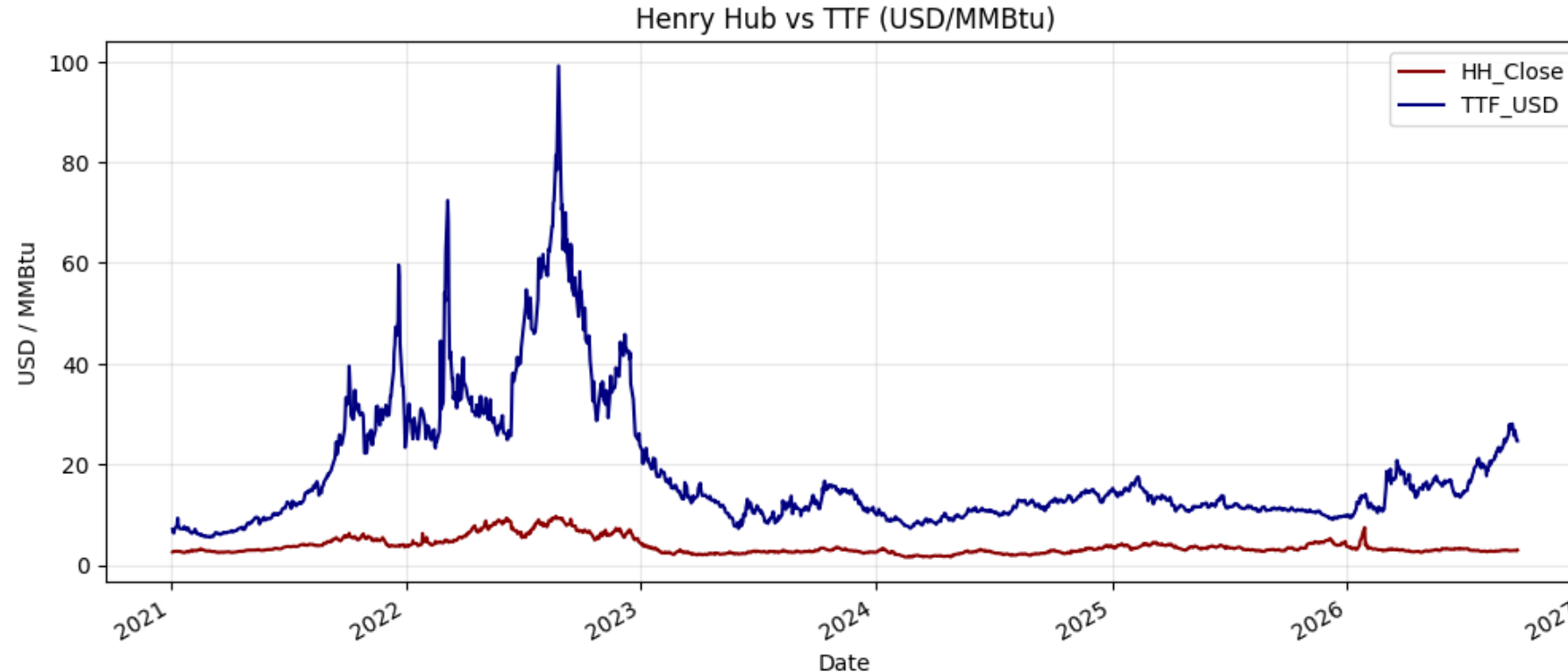
- Three gas benchmarks: Henry Hub (US), TTF (Europe), JKM (Asia).
- TTF converted from EUR/MWh to USD/MMBtu (1 MWh = 3.412 MMBtu, daily EURUSD).
- Aligned on common trading dates: about 1,435 daily observations.
- Transatlantic spread = TTF minus HH. Asia-Europe spread = JKM minus TTF.
- Rough US LNG break-even of about \$4/MMBtu (liquefaction plus shipping) used as the arbitrage test on a 30-day average.

Henry Hub: the US cost base

- Mean price about \$3.71/MMBtu; median \$3.11.
- 2022 spike to a peak of \$9.68 in the crisis regime.
- Then a retreat to roughly \$3, and recent prints near \$2.90.
- Low, stable US gas is the foundation of the export economics.

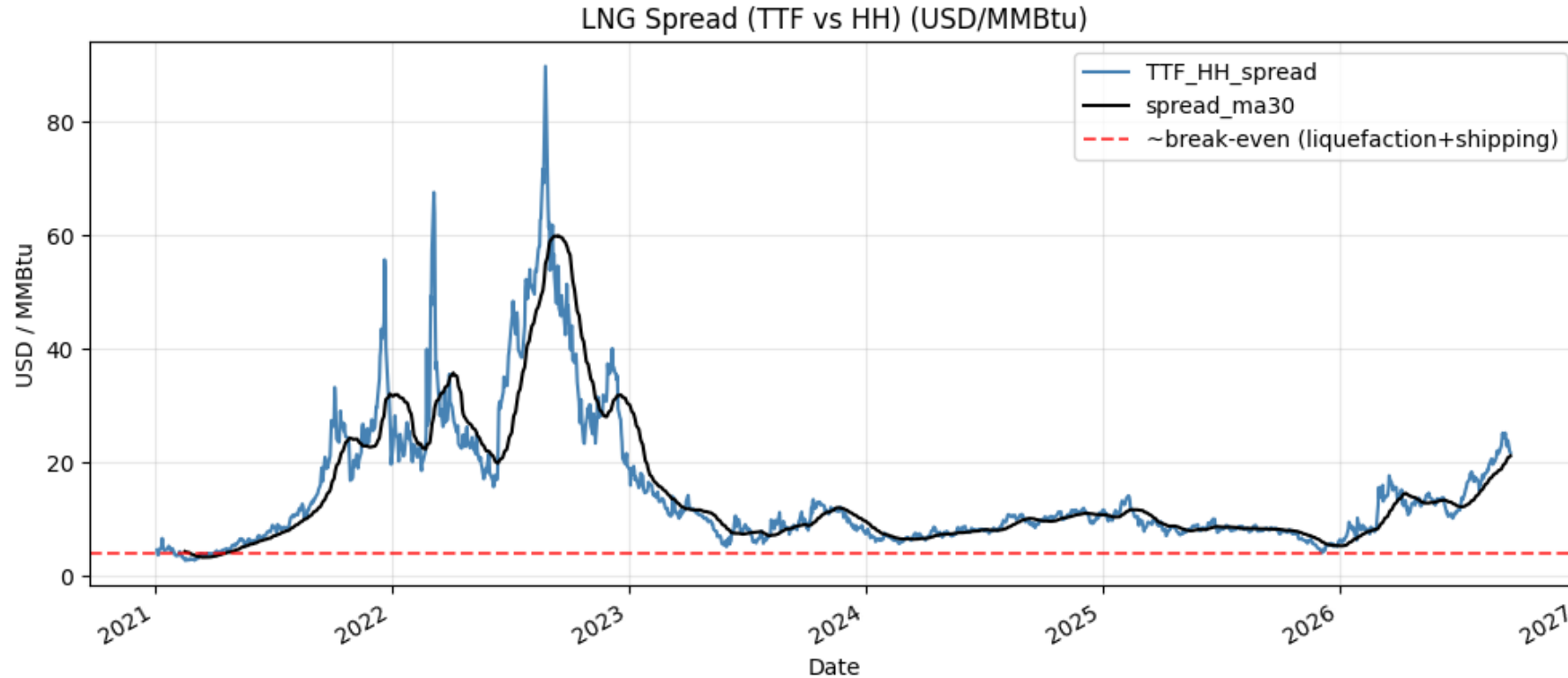


Europe prices at a large premium to the US



- TTF averages \$18.36/MMBtu vs HH \$3.71; TTF peaked at \$99.13 in Aug 2022.
- The gap blew out in 2022, narrowed 2023-2025, and widened again in 2026.

The transatlantic spread covers transport cost



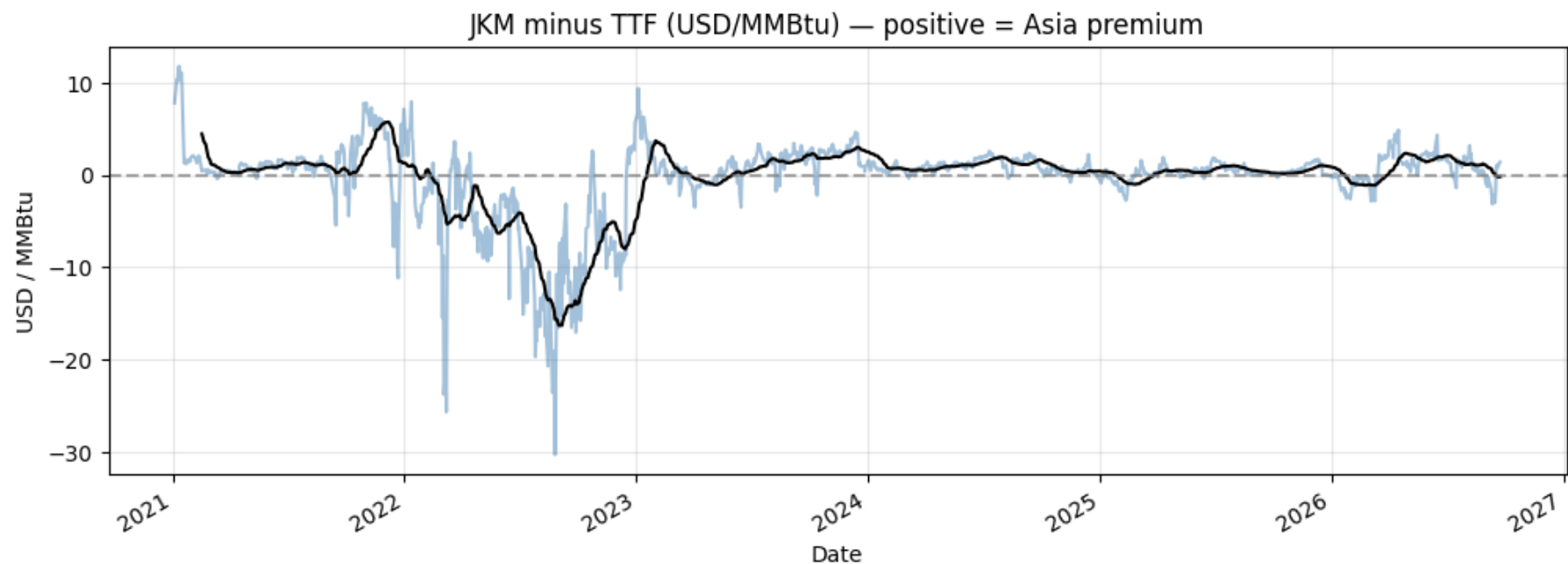
- On a 30-day average the spread sat above the \$4 break-even on 95.3% of days.
- Yearly mean spread (\$/MMBtu): 2022 = 34.1, 2024 = 8.6, 2025 = 8.3, 2026 = 13.7.

Regimes: crisis, normalization, re-widening

Year	Mean TTF-HH spread	Regime
2021	12.6	Pre-crisis tightening
2022	34.1	Supply crisis peak
2023	10.4	Normalizing
2024	8.6	Calm
2025	8.3	Calm (low)
2026	13.7	Widening again

- Aug 2022: daily spread peaked at \$89.83 as Europe rushed to restock.
- Jun 2022 was the weakest 2022 month, and still \$15.65 at its low.

Asia vs Europe: no strong pull either way



Year	2021	2022	2023	2024	2025	2026
Mean JKM-TTF	1.68	-6.57	1.32	0.92	0.32	0.69

- 2022: Europe premium (westward pull). Other years: small Asia premium. Sep 2026 is near zero.

Conclusion

- The US-to-Europe arbitrage was open on about 95% of days, so US LNG has had a persistent margin over the rough \$4 cost.
- The margin is regime-driven: \$34 in 2022, \$8-9 in 2024-25, back up to \$13.7 in 2026.
- Asia and Europe are priced in near balance (JKM-TTF close to zero), so no strong directional pull for cargoes today.
- Takeaway: favor Europe-bound US cargoes while the 2026 spread stays wide; watch JKM-TTF for a shift east.

Caveats: futures-based; the \$4 break-even is a rough single figure; freight, hedging and contract terms are not modeled; JKM futures show zero volume, so treat as indicative.